

MTH166

Lecture-18

Euler-Cauchy Equation

Topic:

Solution of Non-Homogeneous LDE with Variable coefficients.

Learning Outcomes:

To convert equation with variable coefficients (Euler-Cauchy Equation) to an equation with constant coefficients and then solve it with standard known methods.

Euler-Cauchy Equation:

Let us consider 2nd order Non-homogeneous LDE with variable coefficients as:

$$x^2 y'' + xy' + y = r(x) \quad (1)$$

Equation of type (1) called Euler-Cauchy Equation.

$$\text{S.F. : } (x^2 D^2 + xD + 1)y = r(x) \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

The very first job is to convert this equation with variable coefficients to an equation with constant coefficients using an appropriate transformation.

Let transformation be: $x = e^t \Rightarrow t = \log x$

Let $\theta \equiv \frac{d}{dt}$ (Another differential operator)

then $xD = \theta$, $x^2 D^2 = \theta(\theta - 1)$, $x^3 D^3 = \theta(\theta - 1)(\theta - 2)$ and so on...

Equation (2) becomes: $[\theta(\theta - 1) + \theta + 1]y = r(t) \quad (3)$

which is an equation with constant coefficients and we know the methods to solve equ.(3)

Problem 1. Find the general solution of: $x^2y'' + xy' - 4y = 0$

Solution: The given equation is:

$$x^2y'' + xy' - 4y = 0 \quad (1)$$

$$\text{S.F. : } (x^2D^2 + xD - 4)y = 0 \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

then $xD = \theta$, $x^2D^2 = \theta(\theta - 1)$ where $\theta \equiv \frac{d}{dt}$

Equation (2) becomes: $[\theta(\theta - 1) + \theta - 4]y = 0$

$$\text{A.E. : } [\theta(\theta - 1) + \theta - 4] = 0 \Rightarrow [\theta^2 - \theta + \theta - 4] = 0$$

$$\Rightarrow (\theta^2 - 4) = 0 \Rightarrow \theta = 2, -2 \quad (\text{real and unequal roots})$$

$\therefore$ General solution is given by:

$$\Rightarrow y = c_1e^{2t} + c_2e^{-2t} \Rightarrow y = c_1x^2 + c_2x^{-2} \quad \text{Answer.}$$

Problem 2. Find the general solution of: $9x^2y'' + 15xy' + y = 0$

Solution: The given equation is:

$$9x^2y'' + 15xy' + y = 0 \quad (1)$$

$$\text{S.F.: } (9x^2D^2 + 15xD + 1)y = 0 \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

$$\text{then } xD = \theta, \quad x^2D^2 = \theta(\theta - 1) \quad \text{where } \theta \equiv \frac{d}{dt}$$

Equation (2) becomes: $[9\theta(\theta - 1) + 15\theta + 1]y = 0$

$$\text{A.E.: } [9\theta(\theta - 1) + 15\theta + 1] = 0 \Rightarrow [9\theta^2 - 9\theta + 15\theta + 1] = 0$$

$$\Rightarrow (9\theta^2 + 6\theta + 1) = 0 \Rightarrow (3\theta + 1)(3\theta + 1) = 0 \Rightarrow \theta = -\frac{1}{3}, -\frac{1}{3} \quad (\text{equal roots})$$

$\therefore$ General solution is given by:

$$\Rightarrow y = (c_1 + c_2 t)e^{-\frac{1}{3}t} \Rightarrow y = (c_1 + c_2 \log x)x^{-\frac{1}{3}} \quad \text{Answer.}$$

Polling Quiz

The transformation used to convert a LDE with variable coefficients to LDE with constant coefficients is:

(A) $t = e^x$

(B) $x = e^t$

(C) None of these.

Problem 3. Find the general solution of: $2x^2y'' + 2xy' + y = 0$

Solution: The given equation is:

$$2x^2y'' + 2xy' + y = 0 \quad (1)$$

$$\text{S.F. : } (2x^2D^2 + 2xD + 1)y = 0 \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

then $xD = \theta$, $x^2D^2 = \theta(\theta - 1)$ where $\theta \equiv \frac{d}{dt}$

Equation (2) becomes: $[2\theta(\theta - 1) + 2\theta + 1]y = 0$

$$\text{A.E. : } [2\theta(\theta - 1) + 2\theta + 1] = 0 \Rightarrow [2\theta^2 - 2\theta + 2\theta + 1] = 0$$

$$\Rightarrow (2\theta^2 + 1) = 0 \Rightarrow \theta^2 = -\frac{1}{2} \Rightarrow \theta = \frac{1}{\sqrt{2}}i, -\frac{1}{\sqrt{2}}i \quad (\text{complex roots})$$

$\therefore$ General solution is given by:

$$\Rightarrow y = e^{0t} \left(c_1 \cos \frac{1}{\sqrt{2}}t + c_2 \sin \frac{1}{\sqrt{2}}t \right) \Rightarrow y = (c_1 \cos \frac{1}{\sqrt{2}} \log x + c_2 \sin \frac{1}{\sqrt{2}} \log x) \text{ **Ans.**}$$

